linspace ( ) Function

linspace ( ) Function is used to create an array with evenly spaced numbers between a start point and stop point.

Syntax:-

numpy.linspace(start, stop, num=50, endpoint=True, retstep=False, dtype=None, axis=0)

Where,

start – It represents starting element.

stop – It represents ending element.

num – It represents number of parts the element should be divided. Default is 50. It must be non-negative.

endpoint- If True, stop is the last element. If False, stop is not included.

https://www.numpy.org/devdocs/reference/generated/numpy.linspace.html

Creating Array using linspace ( ) Function

Syntax:-

from numpy import *

array_name = linspace(start, stop, num=50, endpoint=True)

Ex:-

from numpy import *

a = linspace(1, 8)

a = linspace(1, 8, num=5)

a = linspace(1, 8, 5)

a = linspace(1, 8, 5, endpoint=False)

1.0	2.75	4.5	6.25	8.0
a[0]	a[1]	a[2]	a[3]	a[4]

Accessing One-D Array Elements

*from numpy import **

a = linspace(1, 8, 5)

print(*a*[0])

print(*a*[1])

print(*a*[2])

print(*a*[3])

print(*a*[4])

1.0	2.75	4.5	6.25	8.0
a[0]	a[1]	a[2]	a[3]	a[4]

Accessing using for loop

*from numpy import **

a = linspace(1, 8, 5)

Without index

```
for el in a:
        print(el)
```

With index

```
n = len(a)
for i in range(n):
        print(a[i])
```

Accessing using while loop

```
from numpy import *
a = linspace(1, 8, 5)

n = len(a)
i = 0
while i < n :
        print(a[i])
        i+=1
```